

■ INTRODUCTION

The CE8303 is a CMOS PFM-control step-up switching DC/DC converter. The PFM controller allows the duty ratio to be automatically switched according to the load, enabling products with a low ripple over a wide range, high efficiency, and high output current. With the CE8303, a step-up switching DC/DC converter can be configured by using an external coil, capacitor, and diode. The built-in MOSFET is turned off by a protection circuit when the voltage at the LX pin exceeds the limit to prevent it from being damaged. This feature, along with the mini package and low current consumption, makes the CE8303 ideal for applications such as the power supply unit of portable equipment.

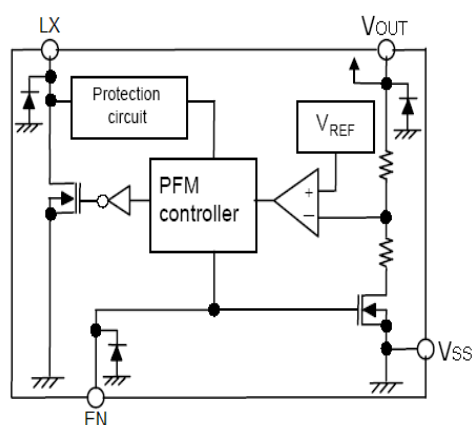
■ FEATURES

- Low voltage operation:
Startup at 0.9 V @ $I_{out} = 1\text{ mA}$
- Working frequency: 300KHz
- External parts: Coil, capacitor, diode
- Accuracy of $\pm 2\%$
- High efficiency: 87% (typ.)
- Shutdown function
- Low ripple, Low noise

■ APPLICATIONS

- Digital cameras
- Electronic notebooks and PDAs
- Portable CD/MD players
- Cameras, video equipment,
- Communications equipment
- Power supply for microcomputers

■ BLOCK DIAGRAM



■ ORDER INFORMATION

● CE8303①②③④

DESIGNATOR	SYMBOL	DESCRIPTION
①	A	Standard LX
	B	Standard EXT
	C	With shutdown, LX
	D	With shutdown, EXT
②③	Integer	Output Voltage (1.8~6.0) e.g.: 3.0V=②:3; ③:0
④	M	Package: SOT-23
	P	Package: SOT-89
	T	Package: TO-92

■ PIN CONFIGURATION

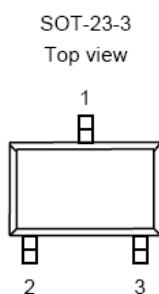


Table 1 CE8303A Series (SOT-23-3 PKG)

PIN NO.	PIN NAME	FUNCTION
1	V _{OUT}	Output voltage pin
2	V _{SS}	GND pin
3	LX	External inductor connection pin

Table 2 CE8303B Series (SOT-23-3 PKG)

PIN NO.	PIN NAME	FUNCTION
1	V _{OUT}	Output voltage pin
2	V _{SS}	GND pin
3	EXT	External transistor connection pin

Table 3 CE8303C Series (SOT-23-5 PKG)

PIN NO.	PIN NAME	FUNCTION
1	EN	Shutdown pin “H”: Normal operation “L”: Step-up stopped
2	V _{OUT}	Output voltage pin
3	NC	(N.C.)
4	V _{SS}	GND pin
5	LX	External inductor connection pin

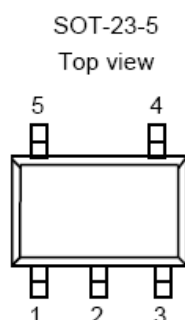


Table 4 CE8303D Series (SOT-23-5 PKG)

PIN NO.	PIN NAME	FUNCTION
1	EN	Shutdown pin “H”: Normal operation “L”: Step-up stopped
2	V _{OUT}	Output voltage pin
3	NC	(N.C.)
4	V _{SS}	GND pin
5	EXT	External transistor connection pin

Table 5 CE8303A Series (SOT-89-3 PKG)

PIN NO.	PIN NAME	FUNCTION
1	V _{SS}	GND pin
2	V _{OUT}	Output voltage pin
3	LX	External inductor connection pin

Table 6 CE8303B Series (SOT-89-3 PKG)

PIN NO.	PIN NAME	FUNCTION
1	V _{SS}	GND pin
2	V _{OUT}	Output voltage pin
3	EXT	External transistor connection pin



Table 7 CE8303C Series (SOT-89-5 PKG)

PIN NO.	PIN NAME	FUNCTION
1	NC	(N.C.)
2	V _{OUT}	Output voltage pin
3	EN	Shutdown pin “H”: Normal operation “L”: Step-up stopped
4	LX	External inductor connection pin
5	V _{SS}	GND pin

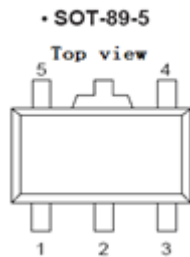


Table 8 CE8303D Series (SOT-89-5 PKG)

PIN NO.	PIN NAME	FUNCTION
1	NC	(N.C.)
2	V _{OUT}	Output voltage pin
3	EN	Shutdown pin “H”: Normal operation “L”: Step-up stopped
4	EXT	External transistor connection pin
5	V _{SS}	GND pin



Table 9 CE8303A Series (TO-92 PKG)

PIN NO.	PIN NAME	FUNCTION
1	V _{SS}	GND pin
2	V _{OUT}	Output voltage pin
3	LX	External inductor connection pin

■ ABSOLUTE MAXIMUM RATINGS

(Unless otherwise specified, Ta= 25°C)

PARAMETER		SYMBOL	RATINGS	UNITS
V _{OUT} pin voltage		V _{OUT}	V _{SS} -0.3 ~ V _{SS} +8	V
EN pin voltage		EN	V _{SS} -0.3 ~ V _{SS} +8	V
LX pin voltage		V _{LX}	V _{SS} -0.3 ~ V _{SS} +8	V
LX pin current		I _{LX}	1500	mA
Power dissipation	SOT-23-X	PD	400	mW
	SOT-89-X	PD	600	mW
	TO-92	PD	500	mW
Operating temperature		T _{opr}	-20 ~+85	°C
Storage temperature		T _{stg}	-40 ~+125	°C
Soldering Temperature & Time		T _{solder}	260°C, 10s	

■ ELECTRICAL CHARACTERISTICS

(Unless otherwise specified, Ta=25°C)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Output voltage	V _{OUT}	—	V _{OUT(S)} ×0.98	V _{OUT}	V _{OUT(S)} ×1.02	V
Input voltage	V _{IN}	—	—	—	6	V
Operation start voltage	V _{ST}	I _{OUT} = 1 mA	—	—	0.9	V
Hold voltage	V _{HOLD}	I _{OUT} = 1 mA	0.8	—	—	V
Current consumption	I _{SS}	V _{OUT} =V _{OUT(S)} +0.5 V	—	7	—	μA
Current consumption during shutdown	I _{SSS}	V _{EN} = 0 V, No load	—	—	1.0	μA
Maximum Oscillation frequency	F _{max}	V _{OUT} = 0.95×V _{OUT} , measure waveform at LX pin	—	300	—	KHz
Duty ratio	Duty		—	75	—	%
Efficiency	EFFI		—	87		%
Current limit	I _{LIMIT}		—	1000	—	mA
Shutdown pin input voltage	V _{SH}		1.5	—	—	V
	V _{SL}		—	—	0.3	V
Shutdown pin input current	I _{SH}		—	—	0.1	μA
	I _{SL}		-0.1	—	—	μA

Remark: V_{IN} =V_{OUT(S)} ×0.6 applied, I_{OUT}=V_{OUT(S)} / 250 Ω

V_{OUT(S)} specified above is the set output voltage value, and V_{OUT} is the typical value of the actual output voltage.

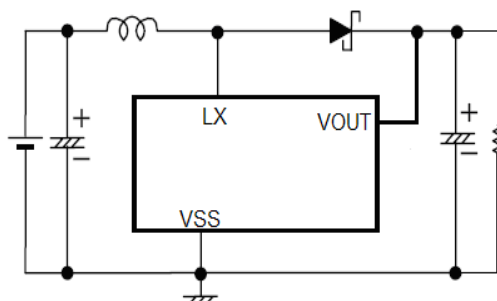
■ STANDARD CIRCUITS

Component: Inductor: 22uH(Sumida)

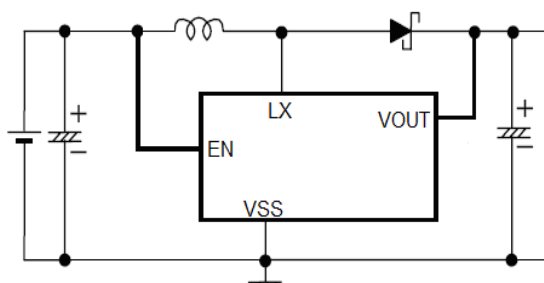
Capacitor: 47uF/10V (Tantalum)

Diode: IN5817、IN5819NMOS: CE2312

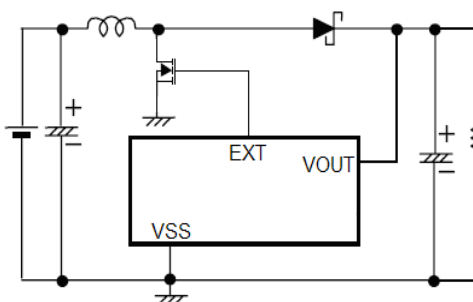
1、CE8303A Circuits:



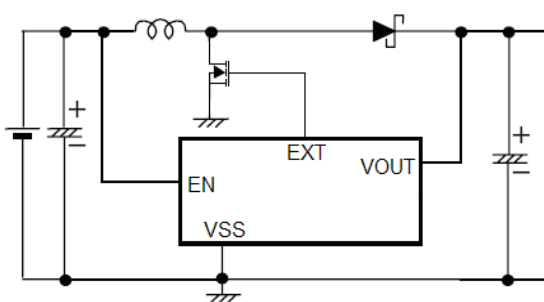
2、CE8303C Circuits:



3、CE8303B Circuits:



4、CE8303D Circuits:

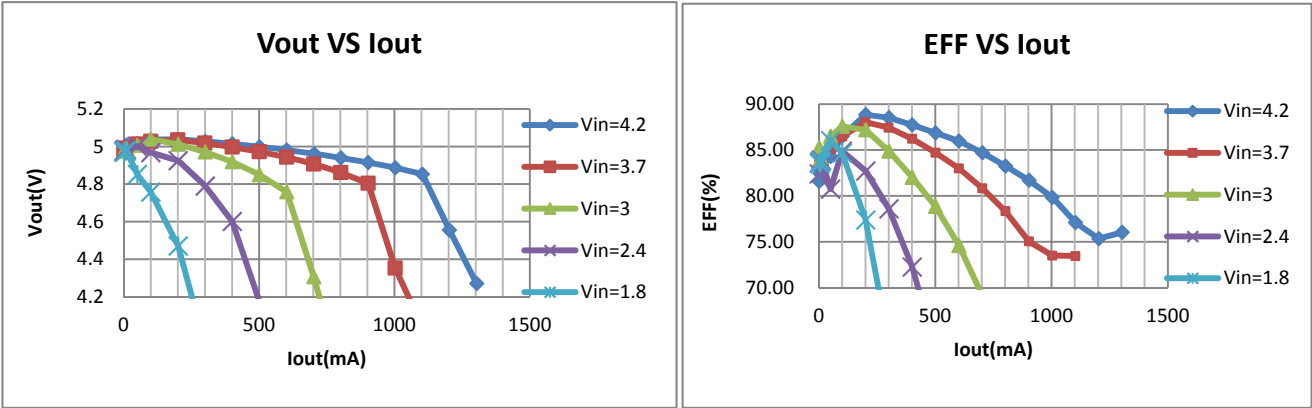


■ TYPICAL PERFORMANCE CHARACTERISTICS

CE8303A50P:

a、V_{OUT} vs. I_{OUT}:

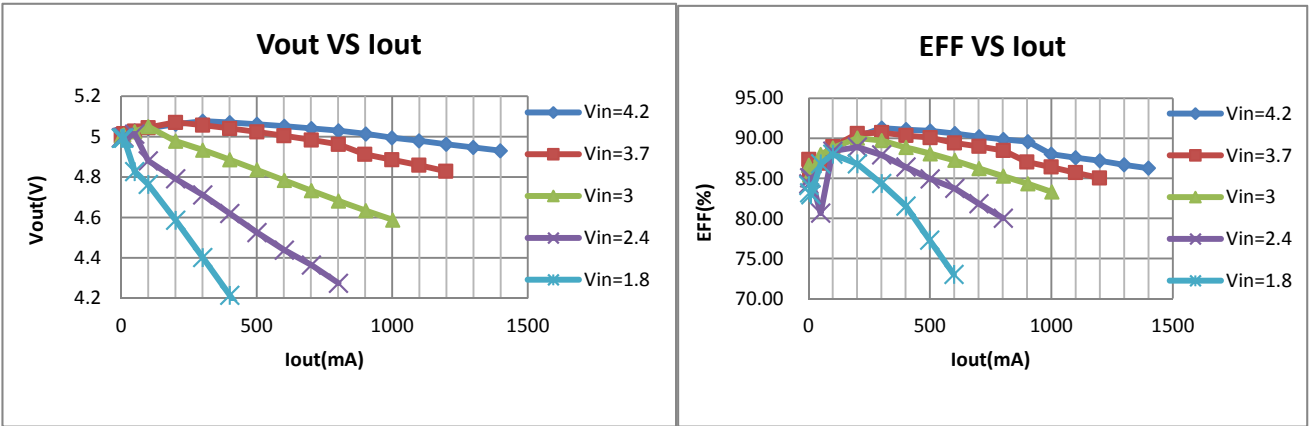
b、Efficiency vs. I_{OUT} :



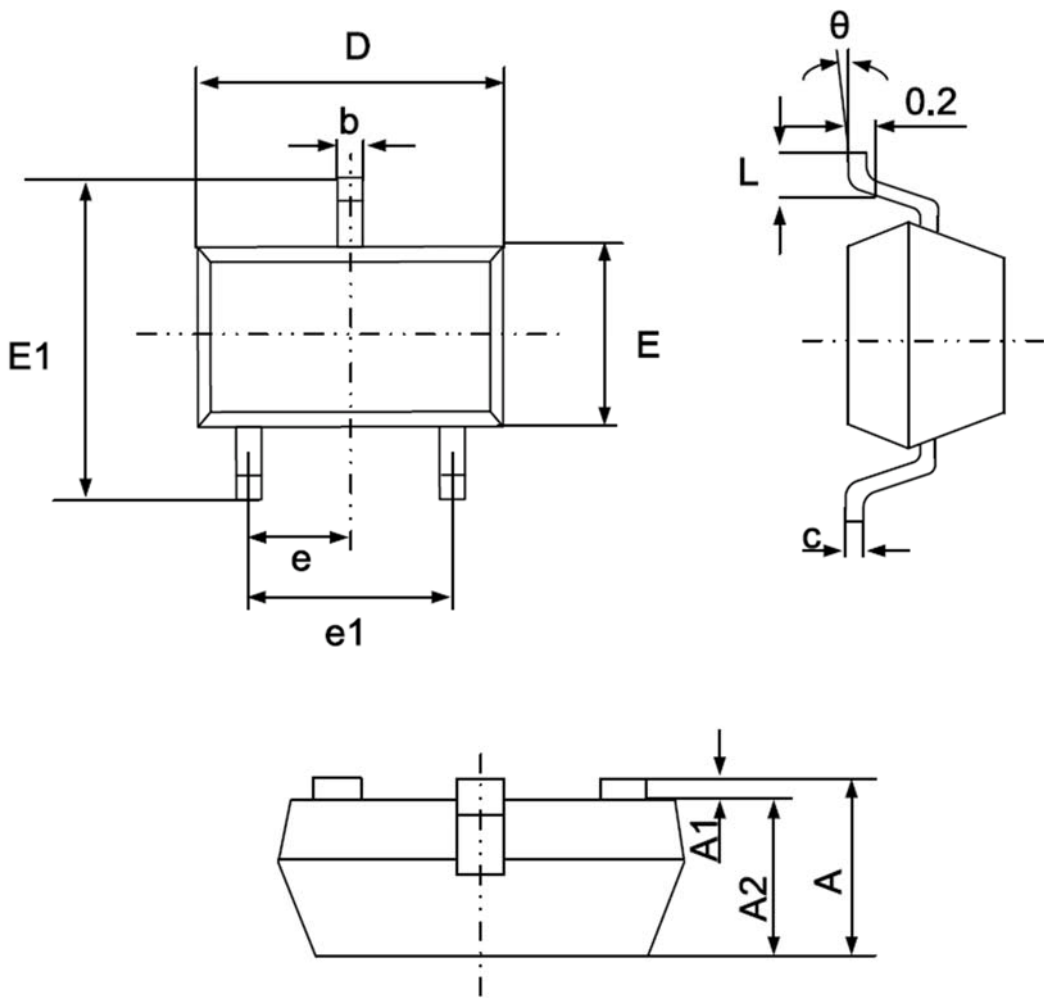
CE8303B50P:

a、V_{OUT} vs. I_{OUT}:

b、Efficiency vs. I_{OUT} :

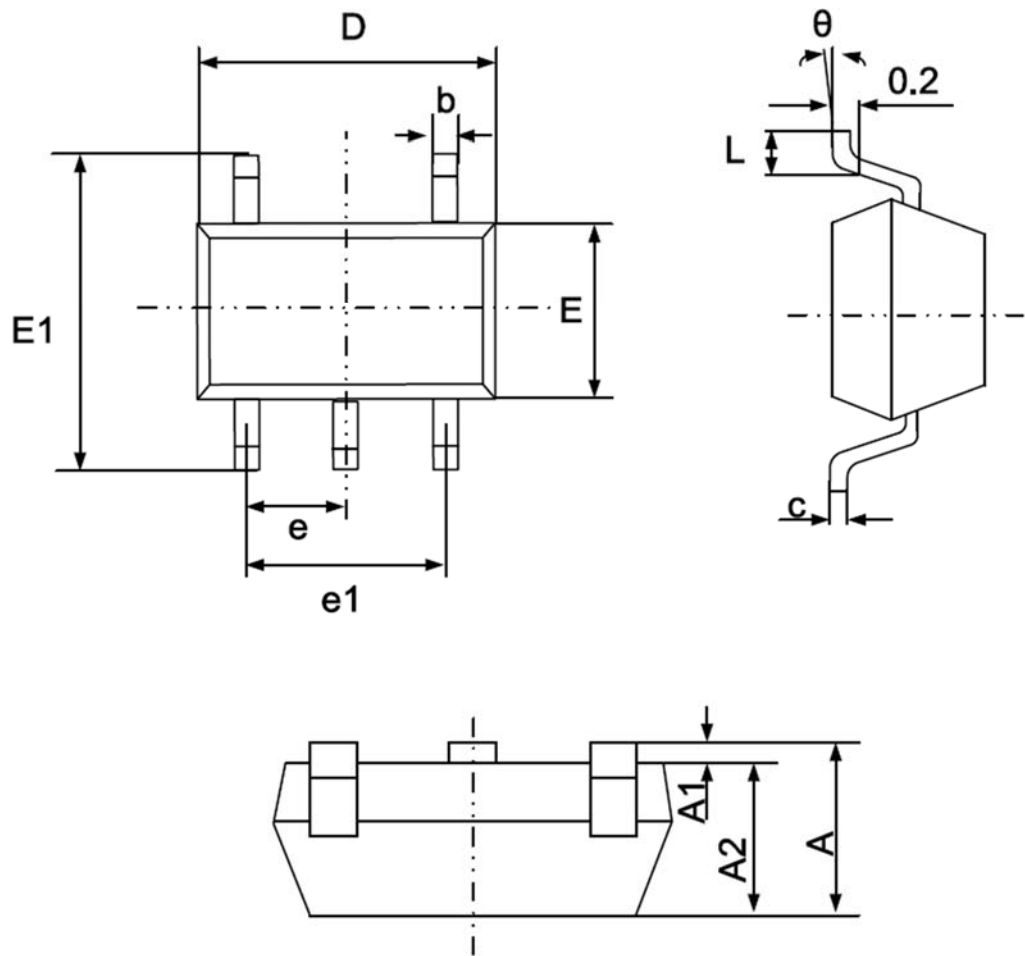


- PACKAGE INFORMATION
- SOT-23-3 PACKAGE OUTLINE DIMENSIONS



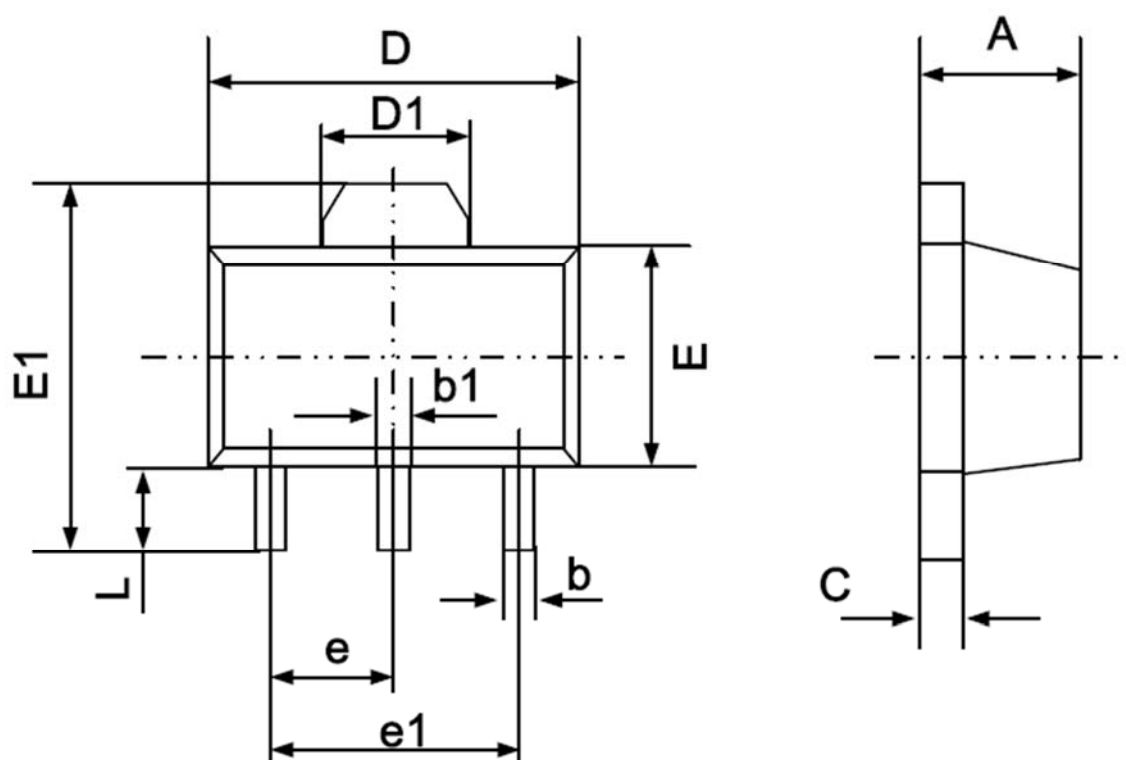
Symbol	Dimensions In Millimeters	
	Min.	Max.
A	1.050	1.250
A1	0.000	0.100
A2	1.050	1.150
b	0.300	0.500
c	0.100	0.200
D	2.820	3.020
E	1.500	1.700
E1	2.650	2.950
e	0.950(BSC)	
e1	1.800	2.000
L	0.300	0.600
θ	0°	8°

● SOT-23-5 PACKAGE OUTLINE DIMENSIONS



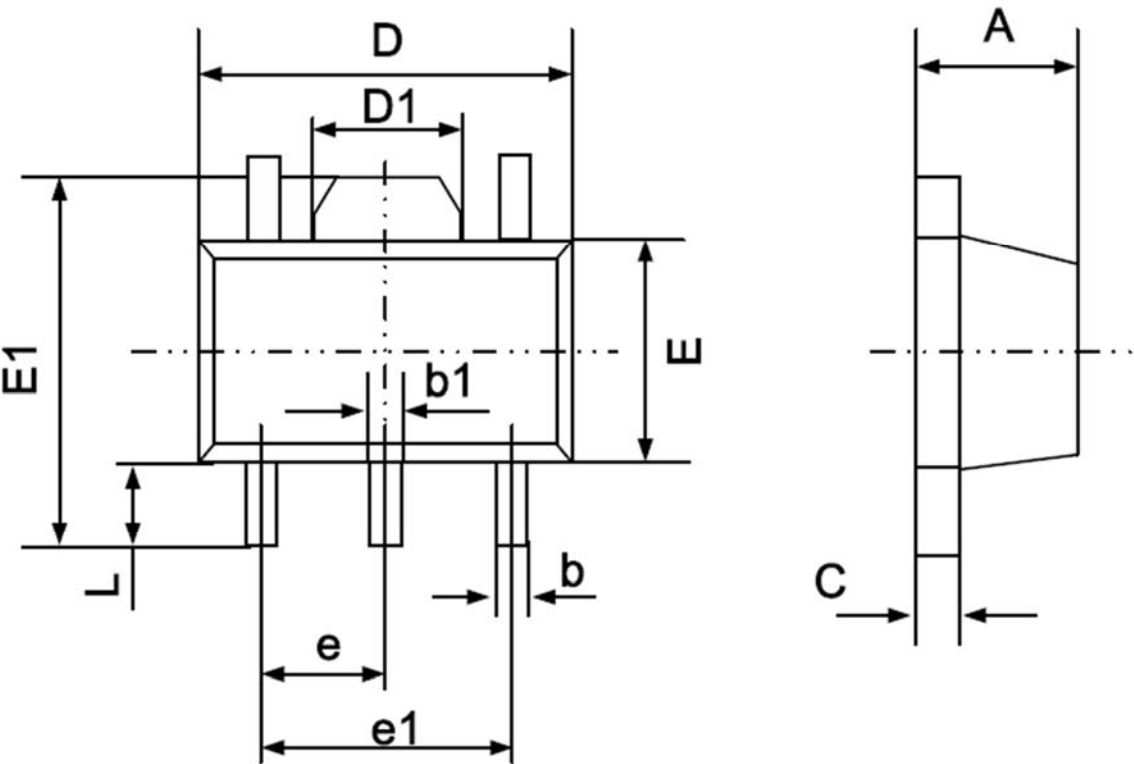
Symbol	Dimensions In Millimeters	
	Min.	Max.
A	1.050	1.250
A1	0.000	0.100
A2	1.050	1.150
b	0.300	0.500
c	0.100	0.200
D	2.820	3.020
E	1.500	1.700
E1	2.650	2.950
e	0.950(BSC)	
e1	1.800	2.000
L	0.300	0.600
θ	0°	8°

- SOT-89-3 PACKAGE OUTLINE DIMENSIONS



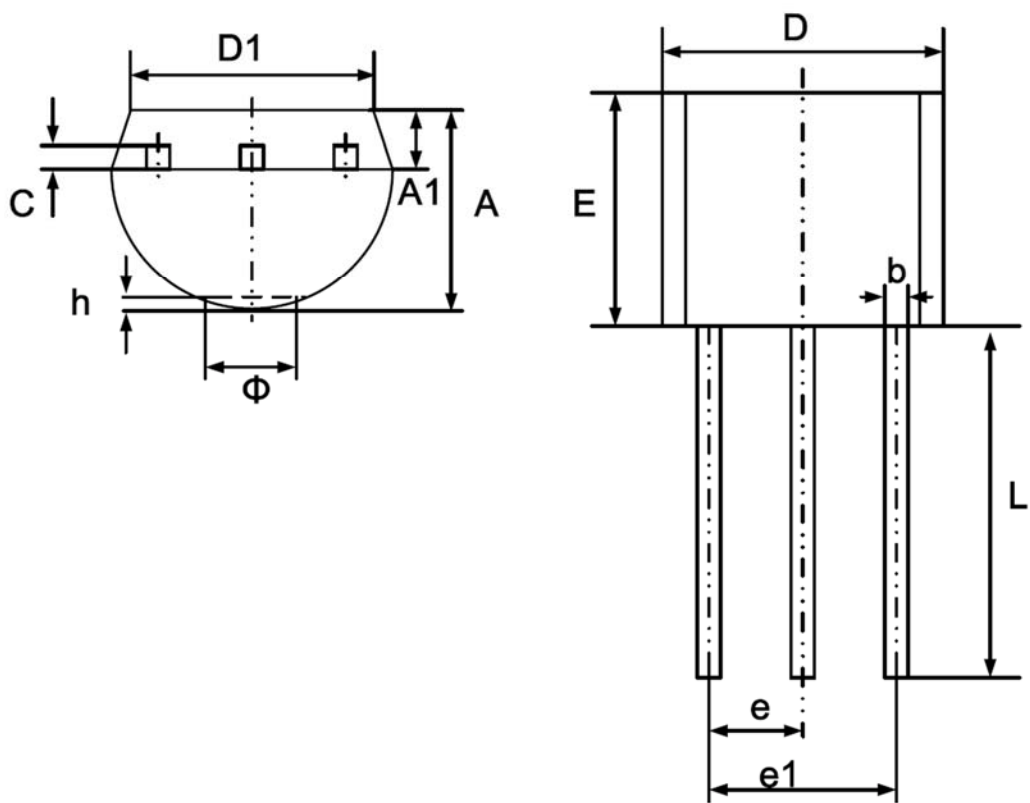
Symbol	Dimensions In Millimeters	
	Min.	Max.
A	1.400	1.600
b	0.320	0.520
b1	0.400	0.580
c	0.350	0.440
D	4.400	4.600
D1	1.550 REF	
E	2.300	2.600
E1	3.940	4.250
e	1.500 TYP	
e1	3.000 TYP	
L	0.900	1.200

● SOT89-5 PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters	
	Min	Max
A	1.400	1.600
b	0.320	0.520
b1	0.360	0.560
c	0.350	0.440
D	4.400	4.600
D1	1.400	1.800
E	2.300	2.600
E1	3.940	4.250
e	1.500 TYP	
e1	2.900	3.100
L	0.900	1.100

● TO-92 PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters	
	Min.	Max.
A	3.300	3.800
A1	1.100	1.400
b	0.380	0.600
c	0.300	0.500
D	4.400	4.800
D1	3.430	
E	4.300	4.700
e	1.270 TYP	
e1	2.440	2.640
L	13. 00	15.00
Φ		1.600
h	0.000	0.380

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